Will Appliance Identification Survive an Indian Grid and a Low-Cost Sensor? A Robustness Study of Current-Waveform Load Signatures

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Abstract

Appliance identification from current waveforms is usually reported on lab-grade recordings, clean electricity supplies and random train/test splits. This article asks whether it survives the conditions of a real Indian deployment: a low-cost clip-on sensor and an imperfect supply. Three public datasets are used, PLAID (USA, 60 Hz), WHITED (eight regions, 50/60 Hz) and HIFDA (Spain, 2025), and every model is evaluated with house- or device-grouped cross-validation, so no test device is ever seen in training. Scale-free waveform features are grouped as time-domain (A), harmonic (B) and voltage-dependent (C), and first-principles models of resistive (R), motor (M) and rectifier-capacitor (E) loads are used to study grid conditions that recordings cannot reproduce. A Random Forest reaches a macro-F1 of 0.81 for 11 appliances and 0.93 for the three load topologies on PLAID. Under simulated sensor degradation, 1 kHz sampling, a 2° current-transformer phase error and a 20 dB signal-to-noise ratio are the breaking points, while 10-bit ADCs and $\pm 5\,\%$ gain error cost nothing. In simulation, a current-only classifier trained on an ideal grid falls from 1.00 to 0.56 macro-F1 once supply voltage distortion reaches 2 %; training across realistic grid conditions removes the drop. On real data, resistive loads copy their own supply's distortion (median harmonic cosine similarity 0.99 on PLAID). On the unseen HIFDA dataset, load topology transfers almost perfectly (macro-F1 0.997) while appliance identity does not (0.21). Waveform shape identifies the circuit inside an appliance; telling apart appliances that share a circuit needs more than one sensor reading.

Keywords

Non-intrusive load monitoring, Appliance identification, Harmonics, Power quality, Current transformer, Low-cost sensing, PLAID, WHITED, HIFDA.

List of Abbreviations

ADC	- Analog-to-Digital Converter	NILM	- Non-Intrusive Load Monitoring
CNN	- Convolutional Neural Network	PF	- Power Factor
CT	- Current Transformer	PFC	- Power Factor Correction
CV	- Cross-Validation	R	- Resistive load
E	- Rectifier-capacitor (electronic) load	RF	- Random Forest
IQR	- Interquartile Range	RMS	- Root Mean Square
kNN	- k-Nearest Neighbours	SNR	- Signal-to-Noise Ratio
M	- Motor / inductive load	SVM	- Support Vector Machine
MCU	- Microcontroller Unit	THD	- Total Harmonic Distortion

1. Introduction

Knowing which appliance draws which current lets a household or a utility see where energy goes without installing a meter on every socket. Non-intrusive load monitoring (NILM) achieves this from a single measurement point [1]. Its core step is appliance identification: naming the device from its voltage and current waveforms. Each load topology, meaning the kind of circuit at the appliance's input, leaves a characteristic signature. A heating element draws a sinusoid in phase with the voltage, a motor draws a lagging current with some magnetic distortion, and a rectifier-capacitor power supply draws narrow pulses near the voltage peak.

Public high-frequency datasets turned this into a benchmark problem. PLAID records plug-level appliances in US homes at 30 kHz [2], and its 2020 release compiles all earlier versions into one reference set [3]. WHITED records household and industrial appliances in Germany, Austria, Indonesia and Canada at 44.1 kHz [4]. HIFDA, released in 2025, records 14 household appliances in Spain at 100 kS/s and has barely been used in published work [5].

Several studies have used these datasets. Gao et al. [6] compared the discriminative power of power, harmonic and binary V–I trajectory features on PLAID and reported 0.85 accuracy with a Random Forest under leave-one-house-out testing. Kahl et al. [10] ranked 36 signatures on high-frequency start-up events and identified promising feature combinations, while Hacine-Gharbi et al. [11] used mutual information to show that odd harmonics carry most of the information for identification. De Baets et al. [7] converted V–I trajectories into images and classified them with a CNN, reaching 0.776 macro-F1 on PLAID; Faustine and Pereira [8] improved this to 0.885 with weighted recurrence graphs. With random record-level 10-fold splits, Himeur et al. [9] reported 0.969 accuracy using local power histograms and an improved kNN. On robustness, Jimenez et al. [12] studied how supply voltage distortion, network impedance and sampling frequency affect steady-state signatures, and a thesis from IIIT Hyderabad [13]

showed that identification models degrade under the 210–250 V voltage fluctuations measured on Indian grids.

Three gaps remain between these numbers and a real deployment in India:

1. **Split protocol.** A random split puts recordings of the same physical device, or the same house, in both training and test sets. Leave-one-house-out results on PLAID cluster near 0.78–0.89 macro-F1 [6–8], well below random-split figures [9].
2. **Sensor quality.** Datasets use lab-grade instruments, whereas a deployable meter would use a clip-on current transformer (CT) and a microcontroller ADC, which add noise, limited resolution and sampling rate, and CT gain and phase errors. These effects have not been swept systematically under grouped evaluation.
3. **Supply quality.** Indian distribution grids show voltage swings of roughly 210–250 V [13] and harmonic voltage distortion. A recording cannot be replayed at a different supply, so the effect of grid conditions on learned signatures is rarely measured.

To overcome these limitations, this paper makes the following contributions: (i) a grouped evaluation of time-domain, harmonic and voltage-dependent features on PLAID and WHITED, at appliance and at load-topology level; (ii) a stress test of five low-cost-sensor effects on real recordings, with their breaking points and an augmentation remedy; (iii) first-principles circuit models of three load topologies, validated against real spectra and used to sweep Indian grid conditions, together with a direct Ohm's-law check on real data; (iv) to our knowledge the first cross-dataset benchmark on HIFDA, using current only, as a clip-on sensor would; and (v) a physics-based explanation of the remaining errors. The rest of the article is organised as follows. Section 2 describes the proposed system, Section 3 the evaluation methodology, Section 4 the results, Section 5 a comparison with existing work, and Section 6 concludes.

2. Proposed System

The proposed system is shown in Fig. 1. Each recording is converted into one feature vector and classified at two levels: the appliance (heater, laptop, fan, . . .) and the load topology (R, M or E). The pipeline (i) loads each record into a common format and detects its mains frequency, (ii) extracts and aligns the last ten steady-state cycles, (iii) computes scale-free feature sets, (iv) optionally degrades the raw recording to imitate a low-cost sensor, and (v) classifies with grouped cross-validation. Circuit simulation and a cross-dataset test run alongside the main pipeline.

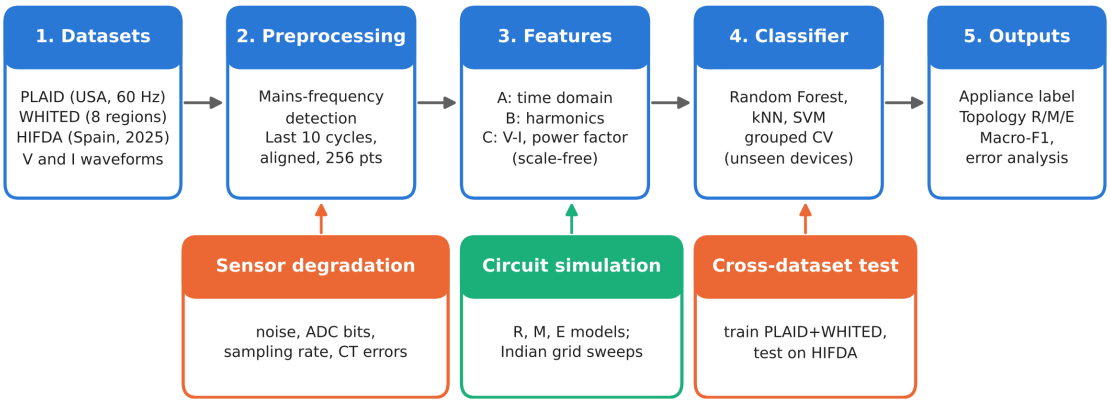


Fig. 1. Proposed system: the five-stage identification pipeline (top) and the three robustness studies feeding into it (bottom).

2.1 Datasets

Table 1 summarises the datasets after loading. HIFDA's voltage channel is band-limited from 300 Hz: about 5 mV was measured at 50 Hz where roughly 325 V is expected, so HIFDA is treated as current-only, like a clip-on CT. Two leakage controls were applied to WHITED: device names were normalised (merging two spellings of one Osram bulb) and two near-identical EcoBay LED entries were merged. WHITED's undocumented region r9 (80 records) was kept after a residual cross-correlation test found no duplicates. Fig. 2 confirms that mains frequency is detected from the data rather than assumed.

2.2 Load Topology Labels

Each appliance class is mapped to one topology, or to X when its circuit is mixed or uncertain. X classes remain in appliance-level experiments and are excluded only at topology level (Table 2).

Table 1. Datasets used in this study.

Dataset	Origin	Records	Classes	Groups	Sampling rate	Mains	Voltage usable
PLAID [3]	USA	1876	16	56 houses	30 kHz	60 Hz	Yes
WHITED v1.1 [4]	Germany, Austria, Indonesia, Canada	1339	56	132 devices	44.1 kHz	50 / 60 Hz	Yes
HIFDA [5]	Spain	2400	12	12 devices	100 kHz	50 Hz	No

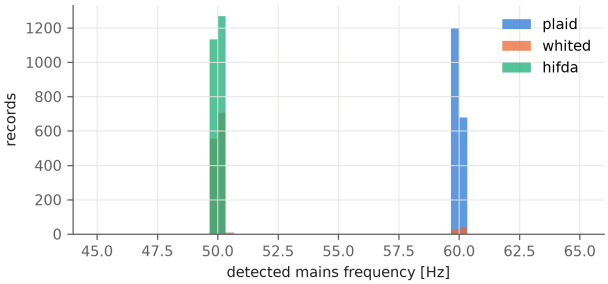


Fig. 2. Detected mains frequency per record. PLAID is at 60 Hz, HIFDA at 50 Hz, and WHITED shows both.

Table 2. Load topologies and the appliances assigned to them.

Topology	Current waveform	Appliances
R, resistive	Sinusoid in phase with voltage	Heater, kettle, incandescent and halogen lamps, iron, coffee maker, hairdryer (heating element dominates; THD 0.01–0.04)
M, motor / inductive	Lagging, slightly distorted sinusoid	Fan, vacuum cleaner, fridge compressor, fixed-speed air conditioner, magnetic-ballast fluorescent tube
E, rectifier-capacitor	Narrow pulses near the voltage peak	Phone and laptop chargers, CFL, LED drivers, monitors, desktop PCs
X, excluded at topology level	Mixed or uncertain	Microwave oven, washing machine, inverter air conditioner

2.3 Preprocessing

Mains frequency is taken as the spectral peak between 45 and 65 Hz of the voltage, or of the current when the voltage has no usable fundamental. The last ten complete cycles are kept, which skips switch-on transients. Each cycle begins at the positive zero crossing of the fundamental and is resampled to 256 points, so the features do not depend on the sampling rate or on 50 versus 60 Hz. The ten cycles are then averaged.

2.4 Feature Sets

All current features are computed on $i/\mathrm{RMS}(i)$, so they describe the shape of the waveform rather than the power rating of the appliance (Table 3). Current THD over harmonics 2 to 40 is given by Eq. (1).

$$\mathrm{THD}_i = \frac{\sqrt{\sum_{k=2}^{40} |I_k|^2}}{|I_1|} \tag{1}$$

The relative phase of harmonic k, $\angle I_k - k\angle I_1$, needs no voltage measurement. It separates harmonics created by the load (core saturation, rectifier pulses) from harmonics that a linear load merely copies from a distorted supply.

2.5 Circuit Models

The supply is modelled with third and fifth harmonics, signed to flatten the voltage peak as on rectifier-loaded low-voltage grids, as given in Eq. (2), where A is chosen so that the RMS voltage equals the target value.

$$v(t) = A\left[\sin\theta + h_3 \sin 3\theta + h_5 \sin(5\theta + \pi)\right], \qquad \theta = 2\pi f t \tag{2}$$

Resistive (R). The current is $i = v/R$ with $R = V_{\mathrm{nom}}^2/P$. The relative current harmonics therefore equal those of the supply, whatever the power.

Motor or inductive (M). A series R–L branch is solved exactly at each supply harmonic k (Eq. 3), and a saturating magnetising branch driven by the flux linkage $\lambda = \int v\,dt$ is added (Eq. 4). The cubic term produces the third harmonic

Table 3. Feature sets.

Set	Needs voltage	Count	Features
A, time domain	No	5	Crest factor, form factor, skewness, kurtosis, peak asymmetry
B, harmonic	No	13	Odd harmonics 3–15 relative to I_1, THD_i, even-harmonic share, relative phase of h_3 and h_5
C, voltage-dependent	Yes	6	V–I loop area, half-wave asymmetry, mid-voltage slope, V–I self-intersections, displacement angle, power factor
P, scale (ablation only)	Optional	1–2	$\log_{10}$ apparent power at nominal voltage; $\log_{10}$ real power

of a saturating iron core.

$$i_k = \frac{a_k}{\sqrt{R^2 + (k\omega L)^2}} \sin\left(k\theta + \phi_k - \arctan\frac{k\omega L}{R}\right) \tag{3}$$

$$i_m = a\frac{\lambda}{\lambda_0} + b\left(\frac{\lambda}{\lambda_0}\right)^3 \tag{4}$$

Rectifier-capacitor (E). A diode bridge (0.8 V per diode) feeds a smoothing capacitor C through a source resistance R_s, with a resistive or constant-power DC load. The capacitor voltage is stepped semi-implicitly at 100 kHz or faster using Eq. (5), and the line current takes the sign of v.

$$C\frac{dv_C}{dt} = \begin{cases} \dfrac{|v| - 2V_d - v_C}{R_s} - i_{\mathrm{load}}, & \text{bridge conducting} \\ -\,i_{\mathrm{load}}, & \text{otherwise} \end{cases} \tag{5}$$

Component values are drawn once at 230 V / 50 Hz from the ranges in Table 4. The same appliances are then re-simulated under each grid condition, so only the grid changes. Fig. 3 shows one simulated appliance of each topology.

Table 4. Component ranges (uniform sampling; 300 training and 150 test appliances per topology).

Topology	Parameter	Range
R	Power P	300–2500 W
M	Power P	40–1500 W
M	Displacement power factor	0.55–0.92
M	Magnetising-to-load peak current ratio	0.10–0.60
M	Cubic (saturation) share of magnetising current	0.20–0.90
E	Power P	5–250 W
E	Capacitance per watt	0.8–3.0 μF/W
E	R_s as a fraction of V^2/P	0.003–0.03
E	DC load type	resistive or constant power

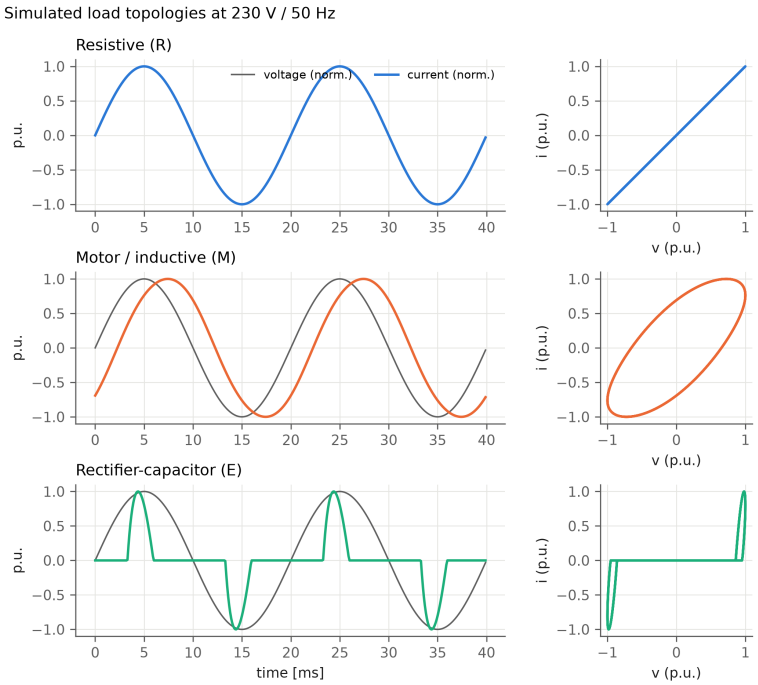


Fig. 3. One simulated appliance per topology at 230 V / 50 Hz: voltage and current over two cycles (left) and the V–I trajectory (right).

3. Performance Evaluation

3.1 Grouped Evaluation Protocol

Every real-data result uses grouped five-fold cross-validation. PLAID is grouped by house and WHITED by physical device (type, brand and region), so a test device is never seen in training. Hyper-parameters are tuned by a grouped inner cross-validation on the training groups only. Only classes present in at least two groups can be evaluated, which leaves 11 PLAID and 28 WHITED appliance classes. HIFDA is never used for training or tuning. Three classifiers are compared on standardised features: distance-weighted kNN ($k \in \{1,3,5,9\}$), a Random Forest (200 trees, balanced class weights) and an RBF support vector machine ($C \in \{1,10,100\}$). The primary metric is macro-F1, the unweighted mean of per-class F1, because the classes are imbalanced; accuracy is also reported.

3.2 Low-Cost Sensor Degradation

Each effect in Table 5 is applied to the raw held-out recording before cycle extraction, one at a time, while models are trained on clean data. A second run trains with random degradations as augmentation. The *breaking point* is the mildest level that costs more than 5 points of macro-F1.

Table 5. Sensor degradation operators.

Effect	Levels tested	Hardware it represents
Additive white noise	SNR 40, 30, 20, 15, 10 dB	Cheap analogue front end
ADC resolution	16, 12, 10, 8 bits	Microcontroller ADC
Sampling rate	10, 5, 2, 1 kHz (anti-alias filtered)	Slow ADC or low-cost meter
CT ratio (gain) error	± 1 %, ± 5 %	Uncalibrated clip-on CT
CT phase error	0.5, 1, 2, 5°	Phase displacement of a CT

3.3 Grid Conditions

The simulated appliances are re-simulated around the Indian nominal supply of 230 V / 50 Hz. Voltage is swept over 210–250 V, frequency over 49.5 and 50.5 Hz, and voltage THD over 2–8 % ($h_3 = \text{THD}$, $h_5 = \text{THD}/2$); a worst case combines 210 V, 49.5 Hz and 8 % THD. *Feature drift* is the shift of a topology's mean feature value from nominal, divided by that feature's standard deviation across all appliances at nominal; a drift of 1 means the class moved by one typical population spread. *Grid accuracy* is the topology macro-F1 of a Random Forest trained either on an ideal grid or on a randomised grid (215–245 V, 49.7–50.3 Hz, voltage THD 0–4 %) and tested under each condition. Simulated

spectra are validated against real ones by the cosine similarity of median odd-harmonic spectra. Because a resistor copies its supply, each dataset is compared with the simulation fed that dataset's measured median supply distortion (Table 6) rather than a clean sine.

Table 6. Measured supply voltage distortion (median over all records).

Dataset	h_3	h_5	Voltage THD (IQR)
PLAID	2.04 %	1.39 %	2.80 % (2.34–3.44 %)
WHITED	1.62 %	1.61 %	2.86 % (2.20–3.36 %)

4. Results Analysis

4.1 Simulation Validation

With a realistic supply, the circuit models reproduce real harmonic signatures (Fig. 4). Cosine similarity is 0.96 / 0.99 for motor loads and 0.98 / 0.98 for rectifier loads (PLAID / WHITED). Resistive loads reach 0.93 on PLAID, up from 0.32 against a clean sine, and 0.73 on WHITED. WHITED stays lower because its supply also carries harmonics 7–15 at levels near h_5 (median h_7 0.91 %) and differs between its eight regions (voltage THD 2.0–4.6 %), while the model's supply carries only h_3 and h_5.

The real data confirms the mechanism directly. For every resistive record, the cosine similarity between its current harmonics and its own voltage harmonics has a median of 0.990 on PLAID (IQR 0.970–0.996, 538 records) and 0.930 on WHITED (IQR 0.792–0.968, 549 records). The "signature" of a resistive appliance is a copy of its local grid's distortion.

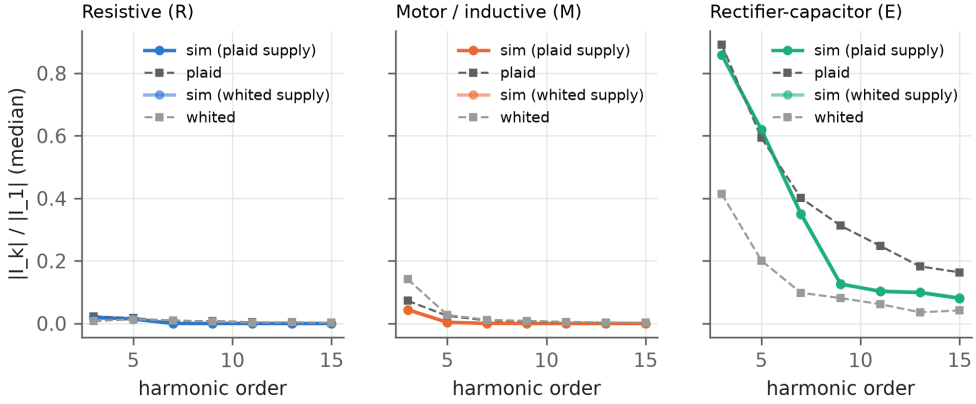


Fig. 4. Median odd-harmonic current spectra of real PLAID and WHITED records against the simulated population fed each dataset's measured supply. Cosine similarity: R 0.93 / 0.73, M 0.96 / 0.99, E 0.98 / 0.98 (PLAID / WHITED).

4.2 Clean Baseline

With the combined feature sets AB and ABC, the Random Forest is the best classifier in every setting, and voltage-dependent features add 0.02–0.10 macro-F1 (Table 7). kNN and SVM trail it by 0.004–0.14 on those sets. WHITED's appliance score is low because most of its 28 classes have only two or three physical devices, so each test device is predicted from one or two others; topology, which pools many devices per class, stays at 0.87. Adding apparent power at the nominal grid voltage (feature P) raises PLAID ABC from 0.806 to 0.833 and WHITED ABC from 0.280 to 0.306.

Table 7. Clean baseline: grouped cross-validation, Random Forest, macro-F1 (accuracy in brackets).

Dataset, level	Classes	A	B	AB (current only)	ABC (with voltage)
PLAID, appliance	11	0.548	0.681	0.702 (0.741)	**0.806** (0.826)
PLAID, topology	3	0.855	0.848	0.869 (0.863)	**0.930** (0.926)
WHITED, appliance	28	0.216	0.237	0.256 (0.358)	**0.280** (0.397)
WHITED, topology	3	0.708	0.821	0.825 (0.843)	**0.868** (0.886)

4.3 Low-Cost Sensor Stress Test

Resolution and gain errors are harmless, while sampling rate, noise and CT phase error break the models (Table 8, Figs. 5 and 6). PLAID topology breaks only at 1 kHz with current-only features (0.87 → 0.65). An 8-bit ADC costs at most 0.04, and ±5 % CT gain error costs nothing, because every feature is normalised by RMS current. CT phase error matters only for the voltage-dependent features: a 5° error drops PLAID appliance ABC from 0.81 to 0.64, below current-only AB at 0.69. Training with random degradations moves the noise breaking point from 20 to 10 dB on both datasets and removes WHITED's topology breaks, at a cost of up to 0.03 clean macro-F1.

Table 8. Breaking points (mildest level losing more than 5 points of macro-F1), clean-trained Random Forest.

Effect	PLAID appliance, AB	PLAID appliance, ABC	WHITED topology, AB / ABC
Noise	20 dB (0.70 → 0.60)	20 dB (0.81 → 0.74)	10 dB / 10 dB
ADC resolution	none	none	none / none
Sampling rate	1 kHz (0.70 → 0.43)	1 kHz (0.81 → 0.73)	none / none
CT gain error	none	none	none / none
CT phase error	none	2° (0.81 → 0.71)	none / 5° (0.87 → 0.72)

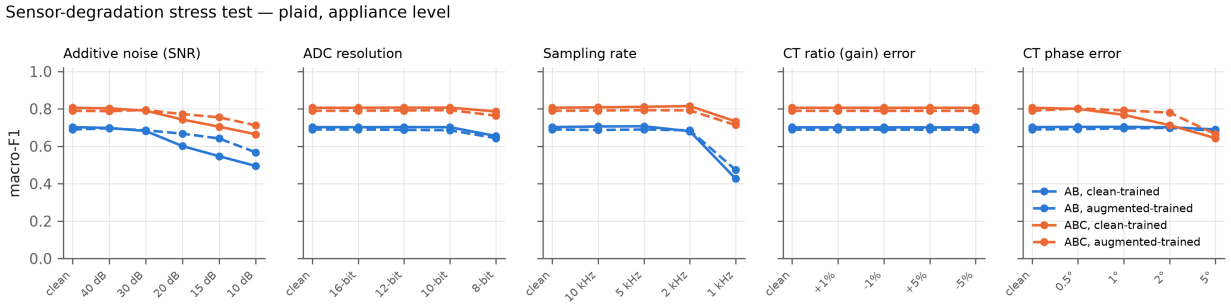


Fig. 5. PLAID, appliance level: macro-F1 under each sensor effect for current-only (AB) and voltage-dependent (ABC) features, clean-trained (solid) and augmentation-trained (dashed).

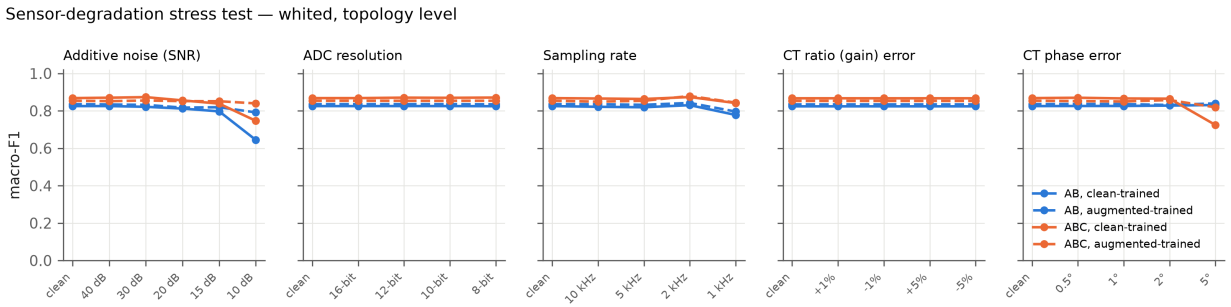


Fig. 6. WHITED, topology level: the same sweep. Noise at 10 dB and a 5° CT phase error are the only breaks.

4.4 Grid Conditions (Simulation)

Voltage magnitude and frequency barely matter; supply voltage distortion is what moves signatures. Across 210–250 V and 49.5–50.5 Hz, drift stays below 0.07 for every topology. At 8 % voltage THD, rectifier loads drift by 5.6 standard deviations in the current-shape features, against 0.25 for R and 0.14 for M (Fig. 7, left), because the rectifier's current pulse sits on the voltage peak that distortion flattens. The consequence for a classifier is sharp (Fig. 7, right). Trained on an ideal grid with current-only features, topology macro-F1 falls from 1.00 to 0.56 at only 2 % voltage THD and stays there, because resistive loads on a distorted supply are mistaken for motors. Adding voltage features, or training across a randomised realistic grid, each restores 1.00 under every condition. For a current-only clip-on meter, only the second remedy is available.

4.5 Unseen Dataset: HIFDA

Trained on PLAID and WHITED and tested on HIFDA with current only, the model identifies the load topology almost perfectly but not the appliance. Topology macro-F1 is 0.997 (recall R 0.992, M 1.000, E 1.000), while

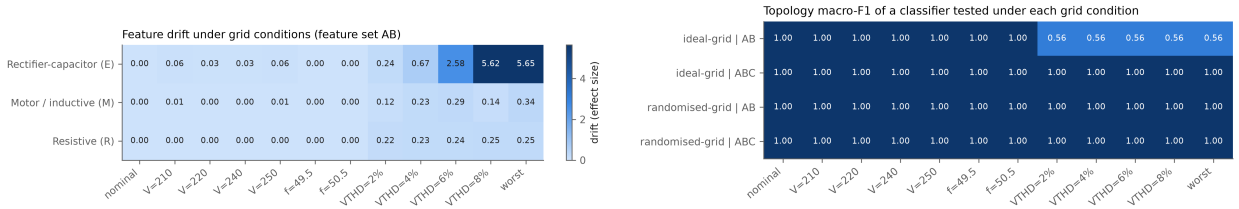


Fig. 7. Left: drift of current-shape features (set AB) per topology under each grid condition. Right: topology macro-F1 under each grid condition for ideal-grid and randomised-grid training, with AB or ABC features.

appliance macro-F1 over the ten shared classes is 0.207 (Fig. 8). Charger, heater and microwave are recognised every time, laptop 29 % of the time, and the other six never. Augmentation leaves both scores unchanged (0.995 and 0.203). The scale feature that helped within datasets hurts here: appliance macro-F1 falls from 0.207 to 0.151, because appliances of one type from different makers draw very different power, while the same circuit always draws the same shape.

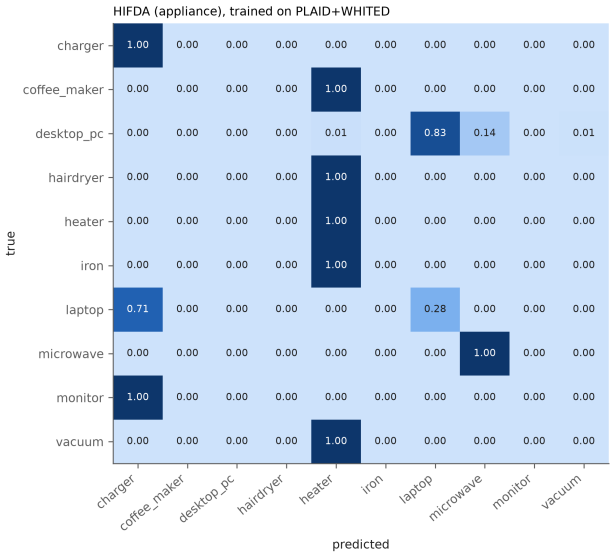


Fig. 8. HIFDA appliance-level confusion matrix (row-normalised); Random Forest trained on PLAID and WHITED, current only.

4.6 Error Analysis

Almost every error stays inside one topology, where only component values separate the classes (Fig. 9). On HIFDA, coffee maker, hairdryer and iron are all predicted as heater, because each is a heating element whose current has the shape of a resistor's. Monitor goes to charger and desktop PC to laptop, all rectifier-capacitor supplies. On PLAID the largest confusions are heater to hairdryer (35 %) and air conditioner to fan or fridge (13–17 %), again same-topology pairs. One error crosses topologies: HIFDA's vacuum cleaner is always predicted as heater. A vacuum uses a universal (series-wound) motor, whose armature and field in series give a high power factor and a near-sinusoidal current at full speed, so it looks resistive.

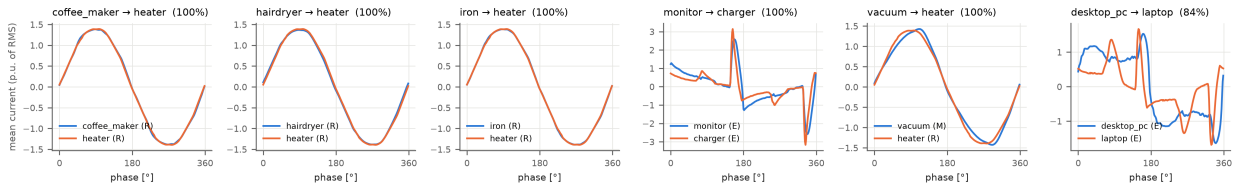


Fig. 9. Mean normalised current cycle of HIFDA's most-confused (true, predicted) pairs, labelled with their topology.

5. Comparative Study

Under the same grouped protocol, the proposed 24-feature Random Forest on PLAID (macro-F1 0.81, or 0.83 with scale) sits inside the 0.78–0.89 band of published leave-one-house-out results, without deep learning (Table 9). The

random 10-fold result of about 0.96 [9] is higher because recordings of the same device appear in training and test. Only rows with the same metric and split protocol are comparable. Our WHITED appliance score is not comparable with earlier WHITED results: those studies assigned WHITED's devices at random to nine artificial houses and scored all 47 classes, whereas we group by real physical device and can evaluate only the 28 classes with at least two devices, most with just one or two training devices. No prior work in Table 9 reports a cross-dataset test, a sensor-degradation study or a grid-condition sweep.

Table 9. Comparison with existing work. Grouped: whole houses or devices held out; random: record-level folds.

Study	Dataset	Method	Split protocol	Metric	Result
Gao et al. [6]	PLAID 2014, 11 classes	Power, harmonic and V–I features, RF	Grouped, leave-one-house-out	Accuracy	0.85
De Baets et al. [7]*	PLAID v1, 11 classes	Binary V–I image, CNN	Grouped, leave-one-house-out	Macro-F1	0.776
Faustine and Pereira [8]	PLAID v1, 11 classes	V–I image / recurrence graph, CNN	Grouped, leave-one-house-out	Macro-F1	0.848 / 0.885
Faustine and Pereira [8]	WHITED v1.1, 47 classes	V–I image / recurrence graph, CNN	Artificial houses (random assignment)	Macro-F1	0.896 / 0.972
Himeur et al. [9]	PLAID, 11 classes	Local power histogram, improved kNN	Random 10-fold	Acc. / F1	0.969 / 0.962
Proposed	PLAID 2018, 11 classes	Shape features ABC, RF	Grouped by house	Macro-F1	**0.806** (0.833 +P)
Proposed	WHITED v1.1, 28 classes	Shape features ABC, RF	Grouped by physical device	Macro-F1	**0.280** (0.306 +P)
Proposed	PLAID / WHITED, 3 topologies	Shape features ABC, RF	Grouped	Macro-F1	**0.930 / 0.868**
Proposed	Train PLAID + WHITED, test HIFDA	Current-only AB, RF	Cross-dataset	Macro-F1	Appliance 0.207, topology **0.997**

*Results as quoted in [8], Table 3.

6. Conclusion

Current-waveform signatures reliably identify the circuit inside an appliance, even with a low-cost sensor, an imperfect grid and data from another country; they do not reliably identify which appliance of that circuit it is. Under grouped evaluation, load topology reaches macro-F1 0.93 on PLAID, 0.87 on WHITED and 0.997 on the unseen HIFDA dataset, while appliance identity reaches 0.81, 0.28 and 0.21. The practical findings for a low-cost Indian meter are as follows.

- **Sensor.** A 10-bit ADC and an uncalibrated CT gain are sufficient. Sampling should be at 2 kHz or faster, the signal-to-noise ratio above about 20 dB, and CT phase error below 1° if voltage features are used.
- **Grid.** Supply voltage distortion, not voltage or frequency swings, is the grid effect that matters. Resistive loads copy it (median similarity 0.99 on PLAID) and rectifier signatures shift the most. A current-only model trained on an ideal grid fails at 2 % voltage THD; training across realistic grid conditions fixes it.
- **Transfer.** Power level helps within one dataset but hurts transfer to new devices (0.207 to 0.151 on HIFDA), whereas waveform shape transfers.
- **Errors.** The remaining confusions are physical: heaters, irons, hairdryers and coffee makers share a resistive shape, chargers and monitors share a rectifier shape, and a universal-motor vacuum cleaner looks resistive.

This study has limitations. The circuit models carry only third and fifth supply harmonics, which explains the lower WHITED resistive match (0.73). Grid effects were simulated rather than recorded on an Indian grid, and only single appliances were studied. Future work should record appliances on Indian feeders with a low-cost CT and microcon-

troller, model the full measured supply spectrum, add power-factor-corrected supplies to the E model, and extend the study to several appliances running at once.

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